

Two-Step Equations in Context

Teacher's Edition

Standard and Targets

MA.7.AR.2.2 Write and solve two-step equations in one variable within a mathematical or real-world context, where all terms are rational numbers.

Learning targets I can solve two-step equations in one variable. I can relate the solution of an equation representing a real-world context to the context.

Essential questions How do you solve two-step equations? How do you interpret the solutions within the real-world contexts they represent?

Building on Lesson 3, and MA.6.AR.2.2 and MA.6.AR.2.3 for one-step equations.

Working toward MA.8.AR.2.1 — multi-step linear equations with rational coefficients, including variables on both sides.

Where this sits in the course Second of five consecutive lessons on MA.7.AR.2.2, required content of course 1205050 that these students are learning for the first time. Unit 1 runs seven grade 7 lessons in a row, and that is the design: **this class owes a whole grade 7 course alongside the grade 8 one**, and the equations strand is where the two meet soonest. MA.8.AR.2.1 — the tested continuation — assumes everything today installs, including the habit of ending on a sentence rather than a number.

Materials Whiteboards and markers. Calculators. Worksheet 7A.01.04. The properties-of-equality anchor chart from Lesson 3, still on the wall.

Lesson at a Glance

Windy Hill periods run **53 minutes**.

Segment	Slides	Time
Settle in; post the learning target	1	1 min
Warm-Up: same context, different question	2	4 min
Notes I–IV: define the letter, solve-check-interpret, dividing first, a negative answer	3–6	13 min
Example 1: the fence	7–9	5 min
Example 2: a runner's plan	10–12	5 min
Example 3: three ways to lose a correct method	13–15	5 min
Whiteboards: 9 questions, each with its reveal	16–33	15 min
IXL — start today's skills	34	5 min
Total		53 min

Where the lesson lives: Slides 13–15. All three rows are errors your students are about to make, and row 3 is the error the whole of Lesson 5 is built around.

This lesson fits, unlike the two before it. Math Nation's components total about 45 minutes, which leaves room for notes. The thirteen minutes of notes are where the interpret habit gets installed, and that habit is the only genuinely new thing today — the solving is Lesson 3 again.

No partner or group work anywhere in this lesson. Math Nation's version has five: the component titled “Collaboration,” a “check your work with your partner,” a “discuss the similarities and differences with your partner,” a “work with your partner... work together to complete the table,” and a Think-Pair-Share in the notes. Wilma and Maurio's two methods are now Notes III, taught from the front, and the three-error table is slides 13–15.

Vocabulary

variable — the quantity in a situation that changes

constant — the quantity in a situation that stays the same

coefficient — the number multiplied by the variable

solution — the value of the variable that makes the equation true

A note on the glossary. Math Nation lists no key terms for this lesson and five additional terms, all names of the properties of equality — those live on the Lesson 3 anchor chart. The four terms above are the ones this lesson actually turns on, because every context question asks students to tell the changing quantity from the fixed one.

Slide-by-Slide Teaching Notes

Slide 1. Two-Step Equations in Context 1 min

- Post the learning target.
- The mathematics did not change overnight — what changed is that the number now has to be interpreted, and an unlabeled number is not an answer.
- Say that at the start and hold to it all lesson.

Slide 2. Warm-Up 4 min

- TYPE: SSDD — same context, different deep structure.
- INDEPENDENT.
- ANSWERS: 1) $45 + 18(6) = \$153$, arithmetic only. 2) $45 + 18m = 243$, so $m = 11$ months. 3) 45 is a one-off charge; 18 is charged every month. Question 1 needs no equation at all and question 2 does, and noticing that difference is the skill. Question 3 is Notes I arriving before it is taught.
- OFF THE SLIDE, YOURS TO SAY: One situation. Three different questions, and only one of them is an equation.

Slide 3. Notes: I. Say what the letter means before you solve 3 min

- THE NEW REQUIREMENT IN THIS LESSON.
- The panel is the standard the worksheet key applies, so say it now and mean it.
- A student who writes only a number has done the mathematics and not answered the question, and that distinction is worth points on the FAST.
- OFF THE SLIDE, YOURS TO SAY: “ $t = 15$ ” is not the answer. “The monthly fee is \$15” is the answer.

Slide 4. Notes: II. Solve, check, interpret — every time 3 min

- Three things, and students reliably do the first.
- Write the three words on the board and leave them there for the rest of the unit; you will point at them a great deal.
- OFF THE SLIDE, YOURS TO SAY: Skipping the check is how a sign error becomes a wrong answer.

Slide 5. Notes: III. Dividing first works — if you divide everything 4 min

- Both routes are correct and the FAST accepts either.
- Mauro's route is the one students get wrong, because they divide the $-3x$ and forget the 15.6.
- Put the wrong version on the board beside the right one and let them find the missing division themselves.
- OFF THE SLIDE, YOURS TO SAY: Every term must be divided, not just some of them.

Slide 6. Notes: IV. A negative answer is information, not a mistake 3 min

- This is why Notes II's third step is not decoration.
- A student who solves correctly, gets -5 miles, and writes it down without blinking has skipped the only step that would have caught the setup error.
- Whiteboard questions 3 and 4 are this exactly.

- OFF THE SLIDE, YOURS TO SAY: The interpretation step is what catches this. Nothing earlier does.

Slide 7. Example 1 *1 of the 5 min*

- SPOKEN ONLY. w = the width of the fenced part, in feet.
- Three sides means the length once and the width twice — draw it if nobody sees it.
- Notes I before anything else.

Slide 8. Example 1 *2 of the 5 min*

- Standard two-step.
- The $2w$ is the part students miss — one length and two widths, not two of each.
- If the equation is right the arithmetic is easy, which is the point of insisting on Notes I first.

Slide 9. Example 1 *2 of the 5 min*

- Do not let this slide go by in two seconds.
- The sentence is the answer, the units are part of it, and asking whether the sign makes sense is a habit that costs nothing here and saves a whole question in Lesson 7.
- OFF THE SLIDE, YOURS TO SAY: A width, in feet, and positive — so the situation allows it. That check is Notes IV.

Slide 10. Example 2 *1 of the 5 min*

- Whiteboards: the equation only, not the answer. $15 + 6m = 45$.
- The variable is miles, not minutes, and that is the sentence to insist on.

Slide 11. Example 2 *2 of the 5 min*

- Point at $m = 5$ and ask what the 5 counts.
- Miles.
- Not minutes, and not miles per minute.
- Students who answer '5 minutes' have solved the equation and lost the question.

Slide 12. Example 2 *2 of the 5 min*

- Notes IV, made concrete.
- Nothing in the solving catches this; only the interpreting does.
- Ask what a negative distance would mean and let the silence do the work, then say plainly that when the answer is impossible the SETUP is what to re-read, not the arithmetic.
- OFF THE SLIDE, YOURS TO SAY: Half a mile, backwards. The arithmetic is right and the situation is impossible — the warm-up alone is longer than the session.

Slide 13. Example 3 *2 of the 5 min*

- Whiteboards, the middle column only — name the step, not the answer.
- Every one of these students used a legal property.
- Do not let anyone answer with the corrected value yet; the naming is the work.

Slide 14. Example 3 *2 of the 5 min*

- Work each row on the document camera.
- Row 1: the coefficient is -6 , not 6 , so $36 \div (-6) = -6$.

- Row 2: $4y$ means 4 times y , so you divide, and $20 \div 4 = 5$.
- Row 3 is the important one — multiplying both sides by -2 means EVERY term on the right, so $-2(-\frac{1}{2}x) - (-2)(3)$ gives $x + 6$, not $x - 3$.
- Read the coefficient aloud before you start — it is NEGATIVE one half, and the row does not work if you read it as positive.

Slide 15. Example 3 *1 of the 5 min*

- Say the sentence on the slide out loud, because it is the slide.
- Then say that row 3 is tomorrow's lesson arriving a day early, and that the students who can name what went wrong there will find Lesson 5 short.
- OFF THE SLIDE, YOURS TO SAY: Check the step, not just the answer. A legal move on part of an expression is not one.

Slide 16. Whiteboards · Question 1 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 17. Whiteboards · Question 1 of 9 — reveal *about 1 minute with the discussion*

- The baseline, and it is Example 2.
- Everything after this changes exactly one thing about it.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 18. Whiteboards · Question 2 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 19. Whiteboards · Question 2 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: the total, by one.
- The answer stops being a whole number.
- Nothing is wrong — a distance may be 5.17 miles — and students who assume a non-whole answer means an error are the ones to find here.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 20. Whiteboards · Question 3 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 21. Whiteboards · Question 3 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: the sign in the middle.

- The answer is negative, and for THIS situation that is impossible — you cannot run a negative distance.
- Notes IV: go back to the setup.
- Ask what real situation the equation would fit; something that starts at 15 and decreases is a tank draining, not a run.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 22. Whiteboards · Question 4 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 23. Whiteboards · Question 4 of 9 — reveal *about 1 minute with the discussion*

- THE ONE THAT EARNS ITS PLACE.
- One feature changed: what the letter counts.
- Identical equation to question 3, identical arithmetic, and now the negative answer is perfectly sensible — a temperature can fall by 5 degrees.
- So a negative answer is never wrong by itself; it is wrong only against what the letter stands for.
- Notes I and Notes IV, in one pair.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 24. Whiteboards · Question 5 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 25. Whiteboards · Question 5 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: the order of the two terms on the left.
- Same answer as question 1 — the commutative property, and Lesson 3's whiteboard questions 5 and 6 arriving again a day later.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 26. Whiteboards · Question 6 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 27. Whiteboards · Question 6 of 9 — reveal *about 1 minute with the discussion*

- THE ONE THAT CLOSES IT.

- One feature changed: a bracket.
- The characters on the page are almost identical to question 5 and the equation is a different equation — now the 15 is inside, so every one of the six lots includes it.
- Put questions 5 and 6 side by side and ask what the bracket did.
- This is Lesson 5 opening tomorrow.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 28. Whiteboards · Question 7 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 29. Whiteboards · Question 7 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: the letter no longer counts the whole thing. m is the hours AFTER the first three, so the equation answers 4 and the question answers 7.
- THE BOARD TO LOOK FOR SAYS 4, and it is not an arithmetic error — the arithmetic is perfect and the reading is not.
- This is the lesson's own subject arriving in the round: Notes IV says go back to what the letter stands for, and here that is the difference between a right number and a right answer.
- Ask the 4-boards to re-read the first sentence aloud.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 30. Whiteboards · Question 8 of 9 — ask *about 20 seconds of thinking*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 31. Whiteboards · Question 8 of 9 — reveal *about 1 minute with the discussion*

- One feature changed: the letter counts MONEY now, not time, and the 12 is what counts the months.
- THE BOARD TO LOOK FOR SAYS 15 MONTHS, which is the reflex from question 7 firing one question too late — a year is twelve months and the 12 is already in the equation.
- Every number in this question is a plausible answer to some question: 55, 12, 15, 180, 235.
- Only one of them answers the one asked, and the way to tell is to say the units out loud.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 32. Whiteboards · Question 9 of 9 — ask *about 90 seconds of writing*

- Boards down until the cue, all up together, scan the back row first, and question a blank board before a wrong one.
- Do not reveal until every board is up.

Slide 33. Whiteboards · Question 9 of 9 — reveal *about 2 minutes with the closing discussion*

- WRITTEN WORK, and the only question in the round that asks for it.
- With the worksheet gone this is the only place in the period the habit gets built, and IXL takes an answer without ever seeing the work.
- ANSWER: substituting gives 36, not 44.75, so she is wrong; $7x = 44.75 + 4.25 = 49$, so $x = 7$, and $7(7) - 4.25 = 44.75$.
- ALISHIA MADE TWO ERRORS AND THEY BOTH HAVE NAMES, which is why her answer is worth reproducing on the board: she read $4\frac{1}{4}$ as 4.5 instead of 4.25, and she SUBTRACTED it where she needed to add.
- Run her numbers and you land on 5.75 exactly — $44.75 - 4.5 = 40.25$, and $40.25 \div 7 = 5.75$.
- A wrong answer that reproduces is a wrong answer you can teach from.
- Take the fraction error first; the class will find the second one themselves once the first is named.
- COLOR: none on this sequence. One expression and its answer is not two objects and not two slots in a formula, so under rule 5 there is no axis to name and the answer color does the whole job.

Slide 34. IXL *5 min*

- IXL.
- The skills are assigned in Google Classroom, not named here, and they are deliberately not ours -- Croix: "The students need questions that aren't exclusively mine."
- Makes them think." Circulate.
- The thing to watch for is a student answering in the box with no paper in front of them: IXL takes an answer and never sees the work, so this is the one part of the period where the habit can quietly lapse.

Worksheet 7A.01.04 — Answer Key

1. check Frankie	$6(7.25) + 2.5 = 43.5 + 2.5 = 46 \checkmark$ Correct.
2. Frankie's properties	Subtraction property, then division property. Multiplying both sides by $1/6$ instead is the multiplication property and is equally correct.
3. check Alishia	$7(5.75) - 4.25 = 36$, and $36 \neq 44.75$. Not correct.
4. Alishia's errors	Two errors. (1) She subtracted $4\frac{1}{4}$ when she needed to add it. (2) She used $4\frac{1}{4}$ on the left and $4\frac{1}{2}$ on the right. Correct: $7x = 49$, so $x = 7$. Check $7(7) - 4.25 = 44.75$. A student naming only one error has not answered the question, which asks for the errors — worth a follow-up question at the desk rather than the point.
5. gym	$t = 15$. The monthly fee is \$15. Check: $55 + 12(15) = 235$. t is a monthly rate, not a number of months.
6. kayak	$m = 4$. Four ADDITIONAL hours, so 7 hours in all. Check: $24 + 4.50(4) = 42$. The most common wrong answer is “4 hours.”
7. $8 + z/4 = 23$	$z = 60$ Subtraction, then multiplication. Check: $8 + 60 \div 4 = 23$.
8. $5t - 17.2 = 16.3$	$t = 6.7$ Addition, then division. Check: $5(6.7) - 17.2 = 16.3$.
9. $s/3 - 0.6 = 1.2$	$s = 5.4$ Addition, then multiplication. Check: $5.4 \div 3 - 0.6 = 1.2$.
10. $2.4a + 8 = 27.2$	$a = 8$ Subtraction, then division. Check: $2.4(8) + 8 = 27.2$.

Question 4 is the one to circulate for. Alishia's work contains two separate errors and both are needed to produce her answer of 5.75: she subtracted $4\frac{1}{4}$ instead of adding it, and she used $4\frac{1}{2}$ on the right where the left had $4\frac{1}{4}$. Most students find the first and stop. Ask “is that everything?” rather than marking it wrong.

Question 6 is the one students misread. The variable counts hours BEYOND the first three, so $m = 4$ means seven hours at the lake. This is a reading error rather than an algebra error, and it is exactly what “interpret the solution” is meant to catch.

Accept more than one property name throughout. Dividing both sides by 6 and multiplying both sides by $1/6$ are both correct and are different properties. Florida explicitly says not to mandate one path.

Additional Practice 7A.01.04 — Answer Key

Post this in Google Classroom alongside the practice sheet. It is optional work for students who want more.

1. Lisa	Not correct. $8 \div (-2) = -4$, not 4 — she dropped the negative. $x = -4$ Check: $-2(-4) - 3 = 8 - 3 = 5$.
2. Zoey	Not correct. $8 - 5 = 3$, not -3 — the subtraction itself is wrong. $m = 1$ Check: $3(1) + 5 = 8$.
3. taxi	$x = 17$. The trip was 17 miles. Check: $1.75 + 0.75(17) = 14.50$.
4. zipline	$z = 9$. That person crossed 9 lines. Check: $75 + 5.50(9) = 124.50$.
5. runner	$m = 5$. The runner ran 5 miles. Check: $15 + 6(5) = 45$.
6. $125 = 15 + 5x$	$x = 22$ The variable ends on the right. That is the symmetric property, and it is finished.
7. $9.2 = 6.1 + h/6$	$h = 18.6$ Subtraction, then multiplication.
8. $17.5c - 15 = 98.75$	$c = 6.5$ Check: $17.5(6.5) = 113.75$, and $113.75 - 15 = 98.75$.
9. $(t - 50)/4 = 400$	$t = 1650$ The whole numerator is divided by 4, so undo the division first: $t - 50 = 1600$.
10. $16.42 + 2.51r = 26.46$	$r = 4$ Check: $2.51(4) = 10.04$.

Misconceptions to Watch

- **Not realizing that $-x$ means $-1x$.** Named first in Math Nation's list for this lesson, and Example 3's third row. (Math Nation TE)
- **Not knowing what the variable represents in a real-world problem.** Also named in their list. Notes I is the antidote. (Math Nation TE)
- **Not knowing which property of equality to use, or in which order.** Also named. Notes III shows two orders that both work. (Math Nation TE)
- **Applying an inverse operation to only one side.** Florida names this for two-step equations. (Florida B1G-M)
- **Misidentifying which number is the constant and which is the coefficient.** Also named by Florida. The whiteboard sequence surfaces it. (Florida B1G-M)
- **Dividing only some of the terms on a side.** Maurio's method is correct but unforgiving. Notes III (slide 5) and Example 3 (slides 13–15) row 3.
- **Answering with a bare number and no unit.** “15” instead of “the monthly fee is \$15.” Notes I item C.

- **Reading a solution as the total rather than the extra.** Worksheet question 6: $m = 4$ additional hours, not 4 hours in total.

Changes from the Math Nation Lesson

- Every partner and group activity is replaced. There were five, including the component titled “Collaboration.” Wilma and Maurio’s comparison is now Notes III, taught from the front; the three-error table is slides 13–15.
- A teacher tip that named the wrong property was corrected. Math Nation’s note on the Fahrenheit problem says to discuss “why we use the division property of inequality first.” There are no inequalities in this lesson, and the step shown multiplies both sides by 1.8 — that is the multiplication property of equality. Both halves of the phrase were wrong.
- The fence figure was dropped and the wording made to carry it. As printed, Math Nation’s dashed rectangle reads as closed on all four sides, while the equation $48.6 + 2w = 80$ models three; under the four-sided reading the problem has no answer, since $2(48.6) + 2w = 80$ gives a negative width. Example 1 now says “three sides” in the sentence, and the teacher draws it if nobody sees it.
- The kayak problem lost its second person. “You and your best friend go kayaking. The cost to rent a kayak is \$24...” invites a student to double for two kayaks, and the given equation prices one. The friend did no work in the problem.
- The zipline problem on the Additional Practice lost its two names for the same reason — the equation is per person.
- Alishia’s work on the worksheet was left with both of its errors, and the key now names both. As printed, the item asks for “the error(s)” but nothing tells you there are two, and both are required to produce her answer of 5.75.
- The Wrap-Up was replaced. Math Nation’s Cool Down is “Write a summary of what you learned,” keyed “Answers will vary” — the second lesson running with an exit ticket that produces no mathematical evidence. The Ticket Out the Door has students solve, model and interpret.
- Raised-negative notation was normalized. Math Nation writes $\bar{3}x$ in some places and $-3x$ in others; everything here uses the standard minus sign.
- Answer keys were built for both sheets from the same source file as the student versions, so the two cannot drift apart.
- Nothing in the mathematics was changed.